DEVELOPER HANDOFF / V1.0

HEALS ON WHEELS

Referral Workflow & Patient Pipeline Portal

A secure, role-based web application for moving referrals from document intake to clinical review, scheduling, NP visits, notes approval, and billing.

01	Referral Intake	INTAKE SPECIALIST
02	Clinical Review	CLINICAL COORDINATOR
03	Scheduling	OFFICE MANAGER
04	NP Visit	ASSIGNED NP
05	Notes Review	CLINICAL COORDINATOR
06	Ready for Billing	BILLING COORDINATOR

Includes: team login + MFA | Kanban drag-and-drop | AI-assisted referral extraction | staff queues | audit trail | HIPAA safeguards

REFERENCE: VIDEO PROVIDED BY REQUESTOR / SCHEMATIC UI ONLY

01 / PRODUCT OVERVIEW

Product definition and behavior

The source video shows a Kanban-style patient board with columns, counters, colored cards, and quick actions. This specification applies that interaction pattern to the actual Heals on Wheels workflow.

The primary object

A referral case moves through a fixed, auditable six-stage workflow. Each case links to a patient profile, one or more uploaded documents, tasks, appointment records, and billing-readiness checks.

Two independent state dimensions

pipeline_stage records WHERE the case is. work_status records HOW work is progressing. Do not use an ambiguous single color or a single status field for both.

Ownership by role

Each stage has a default owner role and a named assignee. Managers may reassign work; a single clinical coordinator may own both pre-visit and post-visit clinical reviews.

Secure collaboration

Users see role-scoped queues; the office manager sees operational totals across permitted locations. Users can leave internal comments, request documents, and track decisions.

REQUIRED TERMINOLOGY

A case must NOT become "Ready for Billing" simply because the NP visit occurred. It becomes ready only after a qualified reviewer accepts the notes and required documentation is complete.

02 / STATE MACHINE

Workflow contract: referral to appointment

Each stage below has explicit entry, actions, and exit conditions; the backend enforces them regardless of how a card is moved in the browser.

01

REFERRAL INTAKE + ELIGIBILITY

OWNER: Intake Specialist

Entry: referral received through secure upload, fax ingestion, or manual entry.

Required: verify identifiers, referral source, insurance plan / coverage (payer response or documented manual verification), service area, and minimum packet completeness.

Exit: patient/referral record confirmed; coverage outcome recorded; required files attached. Then route to clinical review.

02

PRE-VISIT CLINICAL REVIEW

OWNER: Clinical Coordinator / Reviewer

Entry: verified intake package or approved exception.

Required: licensed/authorized reviewer assesses appropriateness, clinical documentation, urgency, wound information, and need for clarification; records accept / return / reject.

Exit: acceptance decision and reviewer signature / timestamp. Accepted cases route to scheduling; incomplete cases return to Intake / Needs Info.

03

SCHEDULING + ASSIGNMENT

OWNER: Office Manager / Scheduler

Entry: clinically accepted referral.

Required: select state/location, assign NP, confirm patient and provider availability, schedule visit, and record communication attempts and outcome.

Exit: appointment scheduled and acknowledged; awaiting NP visit. Cancellations/reschedules remain in this stage with an audit entry.

02 / STATE MACHINE

Workflow contract: visit to billed

Billing readiness and billing submission are separate events. An auditable "Billed" outcome closes the cycle only after the billing coordinator confirms submission.

04

NP VISIT + DOCUMENTATION

OWNER: Assigned Nurse Practitioner

Entry: appointment scheduled or visit confirmed.

Required: document visit status (completed / no show / rescheduled), actual date of service, clinical note source, ordered services, and any supporting clinical documents.

Exit: visit completed AND documentation submitted. Route to post-visit clinical notes review.

05

POST-VISIT CLINICAL NOTES REVIEW

OWNER: Clinical Coordinator / QA Reviewer

Entry: completed visit with uploaded or linked note.

Required: review note completeness, service documentation, provider authentication, order/signature requirements, and supporting wound/procedure details under policy.

Exit: approve note with reviewer/time or return to NP for correction. Only approved notes advance.

06

READY FOR BILLING -> BILLED

OWNER: Billing Coordinator

Entry: approved post-visit documentation and billing checklist passed.

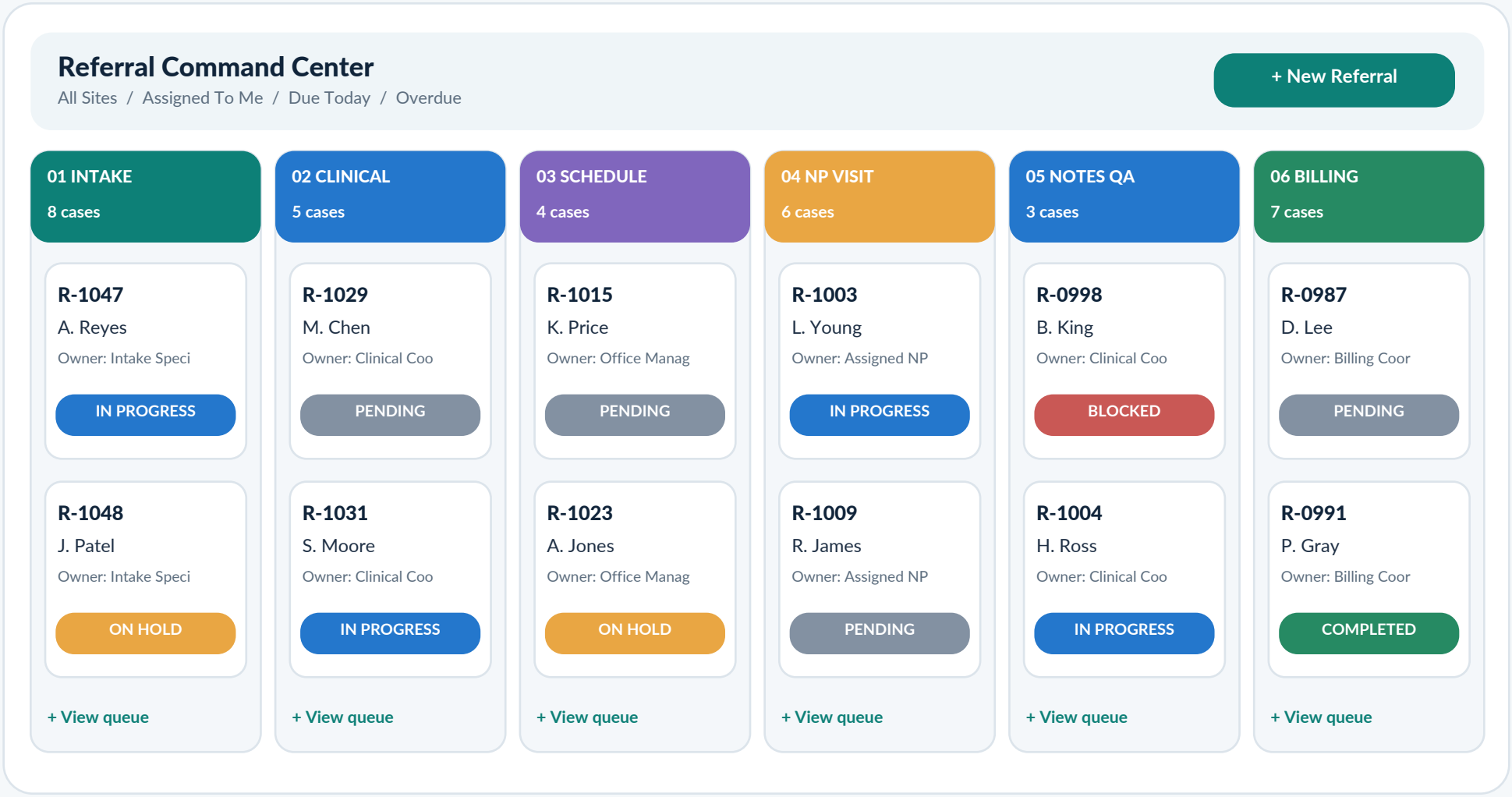
Required: confirm payer/authorization details, DOS, provider, codes, attachments, and claim readiness; prepare and transmit claim in billing system.

Exit: store claim / submission reference and time; set billing_status = SUBMITTED. Rejections and denials remain trackable.

03 / SCREEN SPECIFICATION

Kanban board - interface reference

Synthetic case identifiers are used below. The reference video informs the UI pattern, not the medical data.



DRAG A CARD -> CHECK PERMISSION + STAGE GATE -> RECORD EVENT -> UPDATE COUNTS + ASSIGNEE

03 / WORKFLOW INTERACTION

Status colors, exceptions, and drag rules

Every case displays its stage, work status, assignee, location, elapsed time, and next action. Color never substitutes for a text label or accessibility indicator.

COLOR SYSTEM: WORK STATUS (NOT STAGE)

PENDING	Waiting for stage owner to start.
IN PROGRESS	Actively being worked by the owner.
ON HOLD	Paused with required hold reason and next review date.
BLOCKED / NEEDS INFO	Cannot advance; missing document, eligibility issue, or required correction.
COMPLETED	All mandatory tasks in current stage are finished.

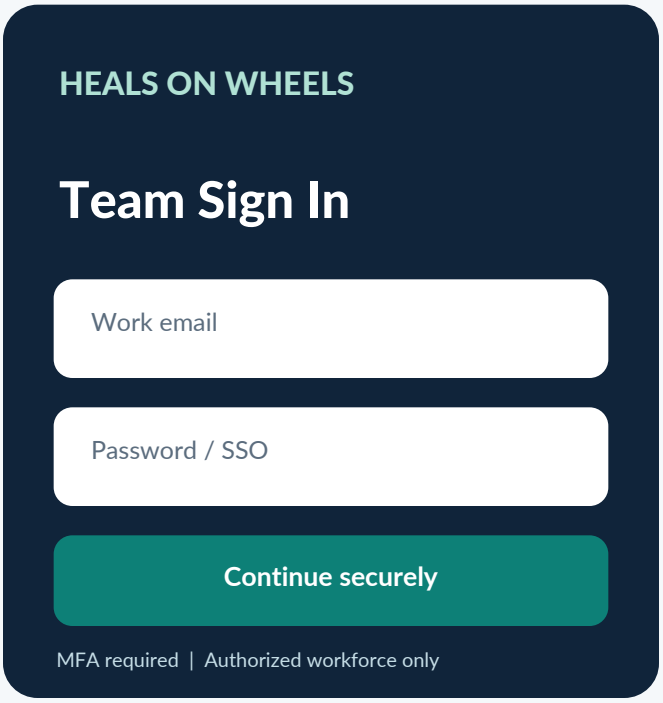
DRAG-AND-DROP CONTRACT

- Only authorized users may drag a case into an allowed next stage. Validate on the server; do not trust client-side rules.
- On drop, show confirmation if it changes the assigned team or exposes a missing mandatory field. Ask for reason on rollback.
- Rejected move returns the card to the original column and explains which requirement is missing.
- Every transition writes an audit event with actor, previous and new stage/status, UTC time, and reason / gate result.
- Handle concurrent editors with optimistic locking or version checks; avoid silent overwrites.

04 / IDENTITY AND PERMISSIONS

Authentication, users, and access control

Invite-only user accounts. Staff members sign in to the same portal but receive role-scoped views and action permissions. Never share a common department login.



ACCESS / ROLE MATRIX

Intake	Intake, coverage, upload, return
Clinical reviewer	Approve / return clinical referral
Office manager	Schedule, assign, supervise queues
NP	Own visits; submit / revise notes
Clinical QA	Review, approve / return notes
Billing	Review readiness, submit claim
Admin / compliance	Provision access, view security audit

ACCOUNT MANAGEMENT AND ENFORCEMENT

- Unique user ID; invite, activate, suspend, revoke, and reset access. Enforce MFA, strong session security, and automatic idle timeout.
- Role-based (RBAC) + location / state / legal-entity scope checks enforced for EVERY API and document retrieval.
- Office manager gets operational oversight; clinical content visibility still requires policy-approved authorization.
- Users can hold multiple roles only if granted explicitly. Record any privileged role or scope change in the audit log.
- Support an NP portal with secure note upload/status and a documented temporary break-glass process only if justified.

05 / DOCUMENT INTELLIGENCE

AI-powered file drop and profile creation

When a referral document is dropped into the intake upload zone, the system creates a DRAFT referral and suggests structured fields. No AI-extracted clinical content is accepted without human review.

01 UPLOAD
Encrypted upload / malware check

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02 EXTRACT
OCR and document classification

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03 REVIEW
Confidence-based human verification

>

04 CREATE
Patient + referral case / queue assignment

FIELDS TO EXTRACT INTO STRUCTURED DRAFT

- Patient: legal name, DOB, phone, address
- Insurance: payer, plan, member ID, group ID
- Referring organization / clinician + contacts
- Referral date, requested service, priority
- Clinical: diagnosis, wound site, relevant history
- Attachments: notes, orders, wound images
- Signature / order presence (not authenticity)
- Document type, source, page and confidence

MANDATORY AI GUARDRAILS

- Use an AI/OCR processor under an appropriate BAA where required; prohibit training on PHI, enforce approved retention, and document subprocessors.
- Display extracted value, source page/snippet, confidence, and editable correction. Route low-confidence, conflicting, and missing fields to staff.
- Check duplicate profiles against normalized identifiers; suggest a match, never auto-merge patient charts without confirmation.
- AI does not determine insurance eligibility, medical necessity, wound diagnosis, coding, or acceptance independently.
- Require staff confirmation before the DRAFT becomes an ACTIVE intake case; maintain provenance and a history of edits.

06 / SECURITY BASELINE

HIPAA security and privacy requirements

Design for HIPAA obligations, not a marketing claim of "HIPAA certification." Actual compliance requires approved vendors, BAAs, policies, risk analysis, workforce training, and ongoing operations.

ACCESS + AUDIT

MFA, unique accounts, role / location least privilege, access reviews, session expiration, audit trails for views, downloads, edits, exports, disclosures, and state changes.

ENCRYPTION + KEYS

TLS for connections; encryption for databases, documents, backups, and keys at rest. Managed key control, secrets management, rotation, and no plaintext PHI in logs.

VENDOR GOVERNANCE

Signed BAAs with cloud hosting, document/OCR/AI services, fax, message delivery, and other applicable PHI-handling vendors. Validate service scope and subcontractors.

SECURE OPERATIONS

Formal risk analysis, incident response / breach assessment, vulnerability remediation, disaster recovery, access termination, workforce training, and periodic audit reviews.

MINIMUM NECESSARY

Show only PHI needed by job function. Avoid PHI in email/SMS notifications, browser analytics, external crash reports, and ordinary developer telemetry.

DATA LIFECYCLE

Document retention/deletion schedules, record integrity, encrypted backups + restore testing, regional/tenant scoping, export controls, and disposal policies.

IMPLEMENTATION GATE

Production use with real patient information is BLOCKED until security review, BAAs, risk assessment, test evidence, and required operational policies are approved by responsible leadership / compliance counsel.

07 / ENGINEERING DESIGN

Recommended technical architecture

A suggested implementation stack, not a vendor mandate. Select platforms offering enforceable security controls and appropriate BAAs. Integration availability must be confirmed independently.

WEB CLIENT	Next.js / React + TypeScript | Accessible Kanban (e.g. dnd-kit) | Role-aware screens
v	
IDENTITY + API	SSO / OIDC + MFA | NestJS or FastAPI | RBAC + tenant/state checks | API validation
v	
WORKFLOW ENGINE	Server-side transition guards | Assignment routing | SLAs / notifications | audit events
v	
DATA + FILES	PostgreSQL | Encrypted object storage | background processing queues | key management
v	
INTEGRATIONS	BAA-covered OCR/AI | clearinghouse eligibility (if contracted) | EHR | billing export

CORE DATA ENTITIES

User, Role, Permission, Site, Patient, ReferralCase, ReferralDocument, InsuranceVerification, ClinicalReview, Appointment, Encounter, NoteReview, BillingCase, Assignment, Comment, Task, AuditEvent, Notification.

DESIGN PRINCIPLE

Document storage and processing must be logically isolated from public assets; every request must be authenticated, authorized, validated, rate-limited as appropriate, and logged without exposing raw PHI.

07 / ENGINEERING DESIGN

API, routing logic, and data contract

The workflow transitions are a backend state machine. The drag-and-drop board is only one client for those transitions; mobile, integration, and batch actions must obey the same rules.

POST /api/referrals/upload

Accept encrypted upload; return document ID and draft-processing status.

GET /api/referrals

Filtered, paged list by role, tenant, site, stage, status and assignee.

PATCH /api/referrals/{id}/transition

Require stage, expectedVersion, action, reason; enforce transition gates.

PATCH /api/referrals/{id}/assign

Assign to authorized role/user within permitted site.

POST /api/referrals/{id}/verify

Approve or correct AI-extracted fields with provenance.

POST /api/referrals/{id}/reviews

Record approve / return / reject and qualified reviewer identity.

POST /api/referrals/{id}/billing

Mark ready, submitted, rejected, or corrected with claim reference.

GET /api/audit-events

Restricted audited access with case, user, date filters.

MANDATORY REFERRAL FIELDS

id, patient_id, tenant_id, site_id, stage, work_status, assigned_role, assigned_user_id, priority, source, received_at, completed_packet_at, stage_entered_at, due_at, hold_reason, hold_review_at, version, created_by, updated_at.

08 / OPERATIONS AND QUALITY

Dashboards, alerts, and edge cases

The office manager needs a single operational dashboard across authorized locations; each other user needs an actionable "My Queue" without excessive PHI exposure.

NEW TODAY Received / unassigned	**WITH INTAKE** Insurance pending	**CLINICAL REVIEW** Awaiting disposition	**TO SCHEDULE** Accepted, unscheduled
VISITS DUE Today / overdue	**NOTES TO REVIEW** Pending corrections	**BILLING READY** Approved, unbilled	**AGING CASES** Breach-risk / held

AUTOMATION RULES

- New referral: assign to intake, display case in queue, notify owner without embedding PHI in unsecured messages.
- On clinical acceptance: assign to scheduler; on confirmed visit: assign to NP; on note submission: route to clinical QA.
- On note approval: mark billing-ready; on failed gate: route back with a named reason and action owner.
- On hold or missing information: require reason, requested party, follow-up date, and visible aging timer.
- Escalation thresholds are configuration, not hard-coded. Example target: 24 hours for intake verification + next 24 hours for scheduling after complete packet.

EXCEPTION FLOWS

Missing documents / Coverage issue / Out of network / Not accepted / Patient declined / No show / Rescheduled / Note correction / Claim rejection / Duplicate / Discharged. Handle as structured outcomes, not hidden or deleted cards; keep the full history and owner.

09 / DELIVERY PLAN

MVP scope and acceptance criteria

The initial release should prove the end-to-end workflow using synthetic test patients before real PHI is allowed. The items below are the minimum definition of done.

RELEASE 1 - CORE OPERATIONS

- [ ] Invite-only sign-in, MFA, role/state/site access permissions, and user administration work end-to-end.
- [ ] Referral uploaded -> AI produces editable draft -> intake validates -> patient/referral case created.
- [ ] Board has six stages, owner counts, colored work statuses, and drag/drop with blocked-move explanations.
- [ ] Scheduling, visit completion, QA approval/return, ready-for-billing, and claim-submitted tracking are testable.
- [ ] Audit events are searchable by authorized reviewers and capture view, change, document access, and transition events.
- [ ] Cases can be searched/filtered, notes and missing-document reasons preserved, notifications sent securely.
- [ ] No PHI appears in public links, browser analytics, logs, unsecured messages, or unauthorized user sessions.

RELEASE 2 - OPTIONAL EXPANSION

- Insurance eligibility clearinghouse integration and structured response capture.
- EHR integration / clinical note sync (subject to access and vendor permission).
- Claims platform integration, rejections and denial follow-up, BI reporting, and mobile NP workflows.

DELIVERABLES: source code, deployment scripts, data dictionary, security tests, runbook, admin training.

10 / HANDOFF NOTES

Open decisions, authoritative references

Before implementation, confirm the decisions below. They are explicit assumptions rather than promises about existing EHR, payer, or billing integrations.

QUESTIONS FOR PRODUCT OWNER

- Which legal entities and state offices share the portal? Is each entity a distinct tenant with isolated PHI?
- Who may approve clinical referrals and post-visit notes? Which licensed roles and supervising policies apply?
- Do NPs chart inside this portal or in an external EHR? Is document linking/manual upload acceptable at launch?
- Which billing product receives claims? Is "Billed" confirmation manual or available via a verified API?
- Which intake sources and payers require integrations first? Who is authorized to confirm eligibility?
- What retention, incident response, SLA, and location-access policies should the system implement?

OFFICIAL HIPAA REFERENCES

HHS - Security Rule overview

HHS - Minimum necessary

HHS - Cloud service provider / BAA guidance

HHS - Privacy Rule overview

SECURITY NOTE: This document is a design specification, not a HIPAA compliance certification.